NAU · DEPARTMENT of MATHEMATICS and STATISTICS

Problem of the Week

#5

Fall 2026

Due Monday, October 26 at 11:59 pm (Arizona time)

Ending in Three

For a positive integer n, look at the last digit of n^n. For example, $3^3 = 27$ ends in 7, and $4^4 = 256$ ends in 6.

(a) Find all positive integers n for which n^n ends in the digit 3.

(b) What fraction of the positive integers have this property? More precisely, let $A(N)$ be the number of integers n with $1 \le n \le N$ for which n^n ends in 3. Find

$$\lim_{N\to\infty} \frac{A(N)}{N}.$$

How to enter

Scan the code, or go to **ac.nau.edu/lms-apps/self-enroll/545026**, to join the POTW Canvas course. Then upload **one PDF** (typed, or neatly handwritten and scanned) to this week's assignment. Open to all NAU undergraduates.

Join on Canvas

The rules

- **Your own work only.** No AI tools (ChatGPT, Gemini, Claude, Copilot, Photomath, …), no internet, no calculators or software, and no help from other people.
- **Explain everything.** A bare answer earns little credit; a clear, complete argument earns full credit.
- Any participant may be asked to explain their solution in person.

Scored 0–4. Full rules, past problems, and model solutions:
naumathstat.github.io/problem-of-the-week
Questions: **ian.williams@nau.edu**